QUESTION 1

What must always be considered when connecting GPIOs across circuits?

Answer is Correct **Score: 1 / 1**

Options

A. Baud rate

Correct

B. Supply voltage levels

C. Crystal frequency

D. Board size

FEEDBACK

Solution: Voltage level mismatches between GPIO and peripheral circuits can cause damage.

Link: https://youtu.be/SE3_88-sN7A?feature=shared&t=17

QUESTION 2

What is the risk of connecting a 5V output to a 1.8V MCU GPIO input?

Not Attempted **Score: 0 / 1**

Options

A. The pin will act like analog

B. LED won't blink

Expected

C. It may damage the GPIO pin

D. It will cause low memory usage

FEEDBACK

Solution: High voltage at the input beyond rated levels can destroy the internal circuitry of GPIO.

Link: https://youtu.be/SE3_88-sN7A?feature=shared&t=74

QUESTION 3

Why is common ground essential in mixed-voltage systems?

Not Attempted **Score: 0 / 1**

Options

A. To synchronize interrupts

Expected

B. To avoid excessive voltage difference across devices

C. To reduce clock jitter

D. To improve baud rate

FEEDBACK

Solution: If grounds are not shared, the reference for voltage levels is undefined, which can cause damaging voltage differences.

Link: https://youtu.be/SE3_88-sN7A?feature=shared&t=141

QUESTION 4

What can happen if the microcontroller sources more current than it can handle?

Not Attempted **Score: 0 / 1**

Options

A. Nothing

B. It will power external devices

Expected

C. Unexpected bugs or dim LED

D. The clock speed increases

FEEDBACK

Solution: If current demand exceeds the GPIO capability, the peripheral (e.g. LED) won't function properly.

Link: https://youtu.be/SE3_88-sN7A?feature=shared&t=286

QUESTION 5

What might happen if a GPIO is forced to sink more current than it can handle?

Not Attempted **Score: 0 / 1**

Options

A. GPIO toggles faster

B. No effect

Expected

C. GPIO and MCU may be damaged

D. Only the ground is affected

FEEDBACK

Solution: Exceeding the sinking capacity can damage the GPIO and internal MCU circuitry.

Link: https://youtu.be/SE3_88-sN7A?feature=shared&t=343

QUESTION 6

What is the purpose of a series resistor with an LED?

Not Attempted **Score: 0 / 1**

Options

A. Limit voltage

B. Convert current to voltage

Expected

C. Limit current to safe levels

D. Delay signal

FEEDBACK

Solution: A current-limiting resistor prevents excess current through the LED, avoiding damage.

Link: https://youtu.be/SE3_88-sN7A?feature=shared&t=582

QUESTION 7

How do you calculate a resistor value for safe LED operation?

Not Attempted **Score: 0 / 1**

Options

A. Using clock frequency and delay

B. By dividing LED voltage by current

Expected

C. Using Ohm's Law on remaining voltage after LED

D. By counting PWM cycles

FEEDBACK

Solution: Subtract LED's forward voltage from supply, then use Ohm's Law (V=IR) to compute resistance.

Link: https://youtu.be/SE3_88-sN7A?feature=shared&t=623

QUESTION 8

What happens if you want to drive a motor needing 100mA with a GPIO that gives 5mA?

Not Attempted **Score: 0 / 1**

Options

A. GPIO boosts its capacity

B. It works fine

Expected

C. Motor doesn't turn or is too slow

D. GPIO pin enters power-saving mode

FEEDBACK

Solution: Insufficient current causes the motor to run too slowly or not at all.

Link: https://youtu.be/SE3_88-sN7A?feature=shared&t=750

QUESTION 9

What's the safe solution to drive high-current devices like motors from GPIOs?

Not Attempted **Score: 0 / 1**

Options

A. Increase PWM speed

B. Use delay functions

Expected

C. Use a transistor as a switch

D. Disable pull-up

FEEDBACK

Solution: A transistor allows the MCU to control high-current devices without directly sourcing that current.

Link: https://youtu.be/SE3_88-sN7A?feature=shared&t=81

QUESTION 10

Why is debouncing necessary with push-buttons?

Not Attempted **Score: 0 / 1**

Options

A. For analog filtering

B. Buttons lose spring action

Expected

C. To eliminate signal noise from bouncing contacts

D. GPIOs switch to analog

FEEDBACK

Solution: Mechanical contacts bounce when pressed, producing unreliable digital signals without debouncing.

Link: https://youtu.be/SE3_88-sN7A?feature=shared&t=963

QUESTION 11

How can debouncing be implemented?

Not Attempted **Score: 0 / 1**

Options

A. Using diodes

B. Only with capacitors

Expected

C. In software by delaying and confirming stability

D. Using voltage regulator

FEEDBACK

Solution: Software waits a few milliseconds and rechecks the pin before confirming the button press.

Link: https://youtu.be/SE3_88-sN7A?feature=shared&t=1040

QUESTION 12

What is Electrostatic Discharge (ESD) and why is it dangerous?

Not Attempted **Score: 0 / 1**

Options

A. Magnetic interference

B. Sudden surge from software bug

Expected

C. High voltage spark from static charge that can destroy circuits

D. Power supply oscillation

FEEDBACK

Solution: ESD from the human body or surroundings can fry delicate microcontroller circuits.

Link: https://youtu.be/SE3_88-sN7A?feature=shared&t=1098